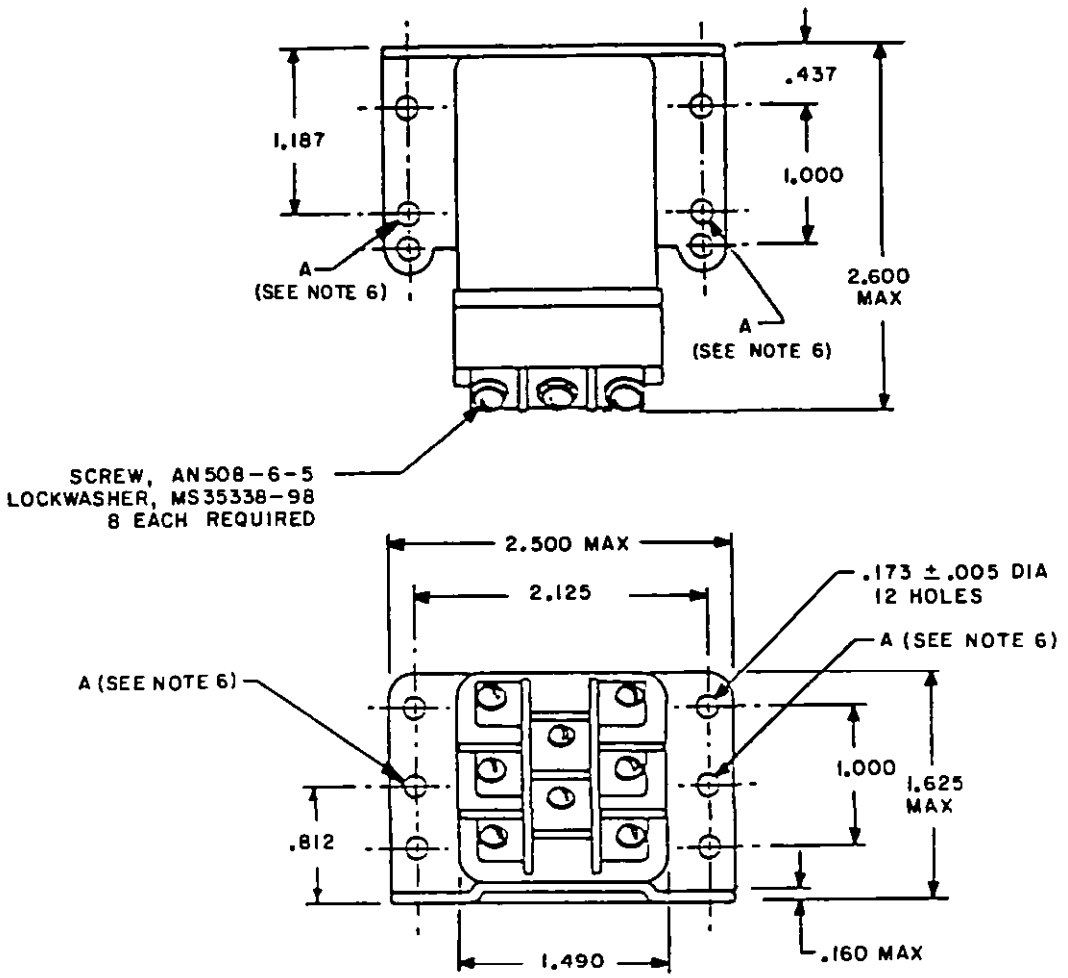
FED. SUP CLASS
5945



Ⓓ ENTIRE STANDARD REVISED

APPROVED 30 Nov 1959 REVISED Ⓓ 5 JUN 87

User activities: Army - Navy - Air Force -

Review activities: Army - Navy - EC Air Force - 11

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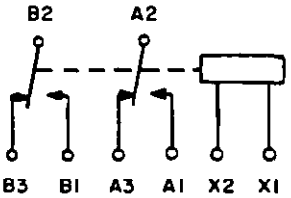
P.A	International interest	TITLE	MILITARY STANDARD
USAF - 85 Other Cust Navy - AS		RELAYS, ELECTROMAGNETIC, 2 PDT 10 AMPERES, TYPE I, HERMETICALLY SEALED	MS24149
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 5

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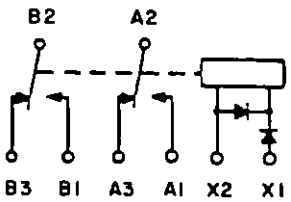
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CIRCUIT DIAGRAM
DC COILS



CIRCUIT DIAGRAM
AC COILS
(SEE NOTES 7 AND 8)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Part number MS24149-D1 replaces part numbers AN3311-1 and AN3311-2.
5. Interchangeability relationship - MS24149 parts can universally replace the inactivated AN3311 parts identified by the same dash numbers; but inactivated AN3311 parts cannot always replace the superseding MS24149 parts. AN3311 is not to be used for new design.
6. Mounting holes 'A' to be used only when replacing AN3311.
7. The use of diodes on ac relays is optional. Actual application must be shown on label.
8. AC coils are inactive for new design.
9. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
10. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

Dash number MS24149-	Type	Coil	Terminal type	Mounting or mating socket	Auxiliary contacts	Max weight in pounds
D1	I	dc	Screw	Bracket	N/A	0.44
A1 1/	I	ac	Screw	Bracket	N/A	0.44

1/ Dash number -A1 is inactive for new design and shall be used for support of existing equipment designs only

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Procurement Specification: MIL-R-6106

P.A: USAF - 85
Other Cust: Navy - AS

International interest:

SUPERSEDES:

TITLE: RELAYS, ELECTROMAGNETIC, 2 PDT 10 AMPERES, TYPE I, HERMETICALLY SEALED

MILITARY STANDARD

MS24149

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TABLE II. Operating characteristics.

MS part no. MS25149-	Coil data												Time - (milliseconds maximum)				
	Coil	Nominal			Max		Max pick-up voltage			Drop-out voltage 2/	Hold voltage 2/	Operate 3/	Release 4/	Contact Bounce			
		Volts 1/	Freq. Hz	Res Ω	Volts	Amperes	Normal 2/	High temp test	Cont current test					Main		Aux	
														NO	NC	NO	NC
D1	dc	28	N/A	160	29	0.25	18	19.5	22.5	1.5	7.0	20	20	2	2		
A1 5/	ac	115	400	N/A	120	0.06	90	95	103	5	30	20	50	2	2		

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ Inactive for new design.

FED. SUP CLASS 5945

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P.A
USAF - 85
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International interest

Procurement Specification
MIL-R-6106

SUPERSEDES:

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MILITARY STANDARD

MS24149

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TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	10	10			10								
Inductive	10	10	10			10								
Inductive														
Motor	50	6	6			6								
Lamp	50	2	2			2								
Transfer load														2/
Mechanical life reduced current	200	2.5	2.5											
Intmd current	50	0.100	0.100											

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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MILITARY STANDARD

MS24149

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FED. SUP CLASS 5945

Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	25 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	10 G
Frequency range	10 - 1,500 Hz
Acceleration	15 G

Electrical characteristics

Minimum insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength (sea level)

	Initial	After life tests
Coil to case	1,250 V rms	1,050 V rms
Aux contacts		
All other points	1,500 V rms	1,250 V rms

Dielectric strength (altitude)

Coil to case		500 V rms
Aux contacts		
All other points		700 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	40 amperes dc
Rupture current	50 amperes dc
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Quality conformance inspection

Performance of groups B and C tests not applicable to -A1.

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